

DATA VORTEX A'26 — ROUND 2

Evaluation Metrics Report

Theme: Rebuilding the Social Engine

Dataset: Labeled Social NLP Training Data

Test Set: 1,822 samples

1. Model Overview

Two NLP classification tasks were evaluated: sentiment classification and topic classification. A transformer-based model was used for sentiment classification, while a TF-IDF based linear classifier was used for topic classification.

Task	Final Model	Features / Configuration
Sentiment	DistilBERT	distilbert-base-uncased, 3 classes, 2 epochs
Topic	Linear SVM	Word + Character TF-IDF, C=2.0, class_weight=balanced

2. Sentiment Classification Evaluation

The final sentiment model was DistilBERT (distilbert-base-uncased), fine-tuned for three sentiment classes: Negative, Neutral and Positive.

Metric	Value
Accuracy	0.6833 (68.33%)
Macro Precision	0.6822
Macro Recall	0.6838
Macro F1	0.6781
Weighted F1	0.6777

Sentiment Classification Report

Class	Precision	Recall	F1-score	Support
Negative	0.7183	0.7701	0.7433	596
Neutral	0.6624	0.5106	0.5767	611
Positive	0.6657	0.7707	0.7144	615

Sentiment Confusion Matrix

Actual \ Predicted	Negative	Neutral	Positive
Negative	459	70	67
Neutral	128	312	171
Positive	52	89	474

Sentiment Error Analysis: The Neutral class has the lowest recall (0.5106). The largest confusion occurs between Neutral and Positive, with 171 Neutral samples predicted as Positive. There are also 128 Neutral samples predicted as Negative. This indicates that neutral social-media language is harder to separate from explicitly positive or negative language.

3. Topic Classification Evaluation

The final topic classifier uses combined word-level and character-level TF-IDF features with a Linear SVM. Class weighting was enabled to reduce the effect of the strong topic-class imbalance.

Metric	Value
Accuracy	0.9308 (93.08%)
Macro Precision	0.9260
Macro Recall	0.5310
Macro F1	0.6160
Weighted F1	0.9157

Topic Classification Report

Class	Precision	Recall	F1-score	Support
Account_Security	1.0000	0.3200	0.4848	25
Community_Discussion	0.9263	1.0000	0.9617	1558
Feature_Feedback	0.7778	0.1167	0.2029	60
Technical_Issues	1.0000	0.6872	0.8146	179

Topic Confusion Matrix

Actual \ Predicted	Account_Security	Community_Discussion	Feature_Feedback	Technical_Issues
Account_Security	8	15	2	0
Community_Discussion	0	1558	0	0
Feature_Feedback	0	53	7	0
Technical_Issues	0	56	0	123

Topic Error Analysis: The dataset has substantial topic imbalance, with Community_Discussion representing the large majority of samples. Although the classifier achieves high overall accuracy, minority-class recall is substantially lower. Feature_Feedback has the lowest recall (0.1167), while Account_Security has recall 0.3200. Many minority-class samples are predicted as Community_Discussion. Therefore, macro F1 is more informative than accuracy alone for evaluating topic performance.

4. Final Evaluation Summary

Task	Model	Accuracy	Macro F1	Weighted F1
Sentiment	DistilBERT	68.33%	0.6781	0.6777
Topic	TF-IDF + Linear SVM	93.08%	0.6160	0.9157

The evaluation demonstrates two different performance characteristics. Sentiment classification remains a more challenging semantic task, while topic classification achieves high overall accuracy but shows weaker minority-class recall because of the dataset's topic imbalance. Macro F1 is therefore reported alongside accuracy to provide a more balanced view of classification performance.

5. Evaluation Methodology

The dataset was split using a group-based strategy based on `post_text`. This prevented exact duplicate texts from appearing in both the training and testing sets. The resulting split contained 7,178 training rows and 1,822 testing rows, with zero exact text overlap between the two sets.

For topic classification, `class_weight='balanced'` was used to account for the severe class imbalance. Evaluation includes accuracy, macro precision, macro recall, macro F1, weighted F1, classification reports and confusion matrices.